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Finite element modeling and modal testing of a wind turbine lattice tower component with interference pin connections

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Abstract

Fatigue failures at fastener holes in structures are undesirable as they can lead to catastrophic mechanical failures. Interference pins create interference fits with joined components to reduce stresses around fastener holes and extend the fatigue life of a structure. In this research, a novel method for finite element (FE) modeling of interference pin connections in a wind turbine lattice tower component was developed. The installation of interference pins was modeled using a two-stage process that causes local stiffness changes in joined members of the component. The local stiffness changes were accounted for in the FE model by using cylinders to represent the interference pins. An experimental setup, including a three-dimensional (3D) scanning laser Doppler vibrometer (SLDV) and a mirror, was used to measure out-of-plane and in-plane natural frequencies and mode shapes of the component. Ten out-of-plane modes and one in-plane mode from the FE model are compared with the experimental results to validate the accuracy of the FE modeling approach. The maximum percent difference between the theoretical and experimental natural frequencies of the component is 3.21%, and the modal assurance criterion (MAC) values between the theoretical and experimental mode shapes are 0.92 or greater, showing good agreement between the theoretical and experimental modal parameters of the component.

Keywords

interference pin, modal analysis, laser vibrometry, finite element modeling, wind turbine lattice tower

1. Introduction

An interference pin, as shown in Figure 1(a), is a type of bolt with a neck that has a larger radius than the radii of the fastener holes (Figure 1(b)) in the members to be joined. The percentage of interference fit commonly studied in the literature is between approximately 0% to 2% (Hu et al., 2019; Kim et al., 2012, 2020a). This type of fastener connection improves the fatigue life of fastener holes (Crews, 1975; Wei et al., 2013) by exerting compressive stresses on the holes (Jiang et al., 2014; Wu et al., 2016). During installation, the pin neck and hole surfaces plastically deform around each other, creating an interference fit. The joined members are then clamped by attaching a nut to the threads of the interference bolt.

FE modeling is a useful tool to simulate and predict dynamic behaviors of structures with interference pin connections. Chakherlou et al. (2009) used FE modeling to predict the effect of an interference pin connection on the fatigue life of fastener holes in a double shear lap joint under a 10 kN cyclic load. Accuracy of the FE results was verified through experimental testing. Residual stresses around a fastener hole after installation of an interference pin are commonly believed to be responsible for the improvement of the fatigue life of a structure (Sun et al., 2016). It has been reported by Song et al. (2015) that installation of interference pins causes stresses to develop around the fastener holes. The stress distribution in the vicinity of an interference pin connection in a structure was recorded from its FE model and compared to experimental results, where good agreement was observed. 3D FE modeling showed that an interference fit hybrid bonded bolted (HBB) connection increased load sharing by 10% when compared to a neat fit HBB connection

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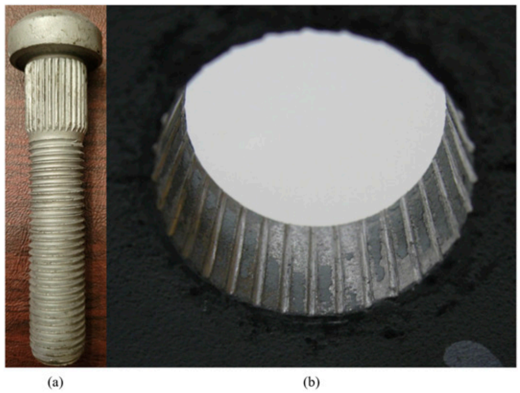

Figure 1. Photos showing the (a) interference pin with splined-like neck, and (b) interference pin fastener hole surfaces.

(Raju et al., 2016). In a separate work (Li et al., 2015), 3D FE modeling was able to predict the type of failure of a single-lap carbon fiber reinforced plastic/titanium alloy structure with an interference fit connection.

In contrast to modeling clearance fit threaded fasteners using RBE2 and zero-length CBUSH elements in FE modeling software (Will and Zhu, 2023), difficulties in modeling interference pin connections arise from contacts of numerous surfaces, and modeling of stiffnesses at the connections (Song et al., 2015). Accurately modeling every contact within an assembly requires fine meshes that can extend the computation time of a simulation (Kim et al., 2013). Another difficulty identified by Kim et al. (2013) comes from local stiffness changes at an interference pin connection, where installation and clamping produces stiffness changes in the vicinity. Including these local stiffness changes in an FE model is needed to accurately predict dynamic behaviors of an assembled structure.

Research has been performed to develop a modeling method where bolted joint connections were replaced with solid cylinders, where the cylinder parameters were meant to emulate local stiffness effects of bolted joint connections (He and Zhu, 2011). The radius of each cylinder was determined from FE modeling of the contact interface between two joined members and the Young's modulus was calculated by treating the material in the clamped region and the bolt as a set of parallel springs. A 3D FE model of a space frame structure was created with cylinders used in place of the bolted joint connections. The first 13 modes of the FE model were compared to the experimental results and the maximum percent error between the natural frequencies was 1.72% and the MAC values between the theoretical and experimental mode shapes were greater than 0.94. Unlike bolted joint connections, interference pin installation causes wear and deformation of joined members, requiring FE modeling of the installation process.

A variation of the FE modeling method described by He and Zhu (2011) was applied to modeling of two aluminum strips joined with six riveted connections (Kim et al., 2020b). To determine the parameters of the cylinders, a two-dimensional FE model was created to simulate plastic deformation of the rivets during the riveting process. The radius and Young's modulus of each cylinder were calculated using methods like those developed by He and Zhu (2011). In a 3D FE model of an assembly of two aluminum strips, six riveted joints were replaced with cylinders; 23 out-of-plane modes were identified for the FE model of joined strips, and when compared to experimental results, the maximum percent error between the theoretical and experimental natural frequencies was 1.63% and the MAC values between the theoretical and experimental mode shapes were all greater than 0.95. In another case, welded joints in a pyramidal truss sandwich panel were simplified by using beam and shell finite elements to replace the original solid finite elements, where the stiffnesses of the beam and shell elements of a welded joint were matched with those of the joint's solid elements (Yuan and Zhu, 2021). However, FE modeling of the installation of an interference pin requires a different method from that of a riveted joint and a welded joint because they deform differently from each other.

A wind turbine can be mounted on a lattice tower, such as that shown in Figure 2, which requires less material to manufacture than other towers, such as those with traditional rolled steel sections, making transport easier and reducing cost (Windpower Engineering and Development, 2015). However, careful consideration is necessary, as failure of a wind turbine lattice tower can occur if the rotation speed of its blades coincides with a natural frequency of the tower (Nezamolmolki and Shooshtari, 2016). The FE modeling method can be used in conjunction with experimental modal analysis data to refine the FE model and accurately predict the dynamic characteristics of a structure, such as a historical building (Standoli et al., 2021), which is a process known as model updating (Monchetti et al., 2023). An objective function is set to search for the optimum solution and aims to minimize the difference between the experimental and numerical modal data (Salachoris et al., 2023). MAC values between the experimental and numerical mode shapes of the structure can be used to assess the accuracy of the structure's FE model (Bianconi et al., 2020).

The goal of this work is to develop a novel FE modeling method for a wind turbine lattice tower component consisting of an L-beam, two flat plates, and four interference pin connections, as shown in Figure 2. The FE modeling software, Abaqus, was used to create a model of the dynamic behaviors of the lattice tower component. To solve the stated challenges with modeling interference pin connections in structures, a model consisting of a two-stage modeling process was used to create cylinders that replace interference pin connections. In the first stage of the process,

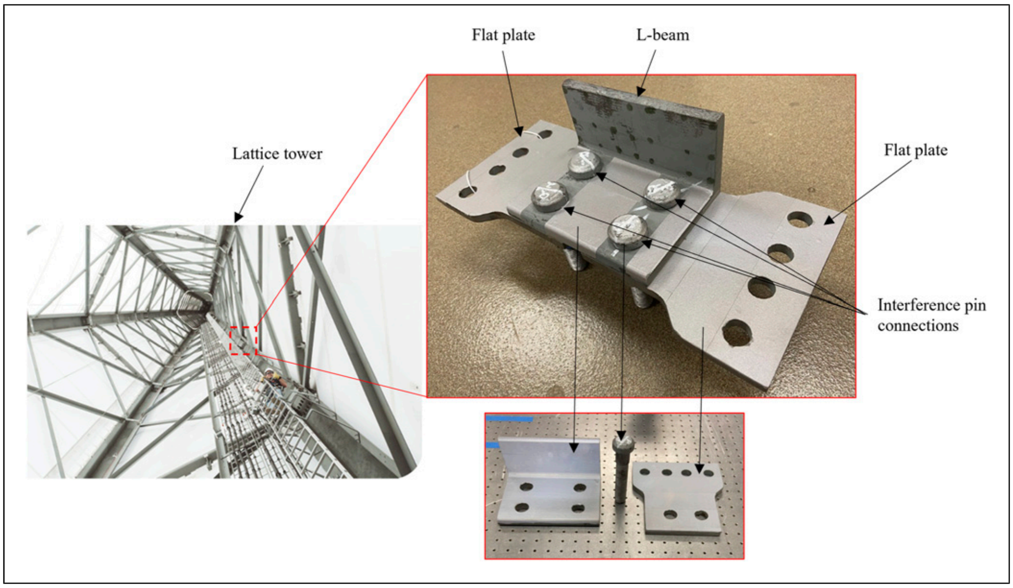

Figure 2. View of the lattice tower (looking up from the bottom) (Windpower Engineering and Development, 2015) with its component consisting of an L-beam, two flat plates, and four interference pin connections.

an FE model simulation of the installation of an interference pin section into an L-beam section and a flat plate section was created. In the second stage of the process, a clamping stress was applied to simulate the clamping force that a nut, torqued onto the threads of the interference pin, would apply to the L-beam and flat plate. Deformation and contact stresses of the joined members from the FE model simulation results were used to calculate the radius and Young's modulus of each cylinder. To calibrate the FE model, the Young's modulus values of the L-beam and flat plate were updated by first determining the natural frequencies of the actual L-beam and flat plate through modal testing using a 3D SLDV. Then, Young's modulus values in the FE model were iteratively varied to minimize the maximum percent difference between the FE model simulation results and the modal test results. The novelty of the proposed FE modeling method is that it determines geometrical and material parameters of a cylinder, which represents the interference pin connection, through simulation of the interference pin's actual installation process. This simplified interference pin connection model can be used in the modeling of the entire lattice tower component. Using this simplified model in the analysis of a large structure can significantly reduce both the computation time and the computer memory requirements, while ensuring that the local stiffness changes inherent in interference fit joints have been accounted for. While effort will be spent tuning the material properties via model updating of the lower-level components in the FE model, namely the L-beam and flat plates, this process ensures accurate modeling results. Natural frequencies and mode shapes of the lattice tower component were experimentally measured using the 3D SLDV and a mirror. Validity of the FE model was assessed using a percent difference formula between the theoretical and experimental natural frequencies and a MAC value calculation between the theoretical and experimental mode shapes. The maximum percent difference between the theoretical and experimental natural frequencies of the component is 3.21%, and the MAC values between the theoretical and experimental mode shapes are 0.92 or greater, showing good agreement between the theoretical and experimental modal parameters of the component.

This paper is structured as follows. The FE modeling methodology is detailed in Section 2. FE simulation of the installation of an interference pin is discussed in Section 2.1. Model updating of the Young's modulus values of the L-beam and flat plate is described in Section 2.2, while FE modeling of the lattice tower component is described in Section 2.3. The experimental setup used to measure natural frequencies and mode shapes of the lattice tower component is detailed in Section 2.4. Natural frequencies and mode shapes of the FE model of the component and their experimental results are compared in Section 3 and a discussion of results is presented there. Finally, some conclusions of the work are given in Section 4.

2. Methods

2.1. Simulating interference pin installation

To simulate the installation of an interference pin, a 3D FE assembly model of a section of the interference pin, L-beam, and flat plate was created using Abaqus. Sections were used because they require fewer mesh elements to model, decreasing the simulation computation time. The

Table 1. Material properties of the interference pin, L-beam, and flat plate.

Part	Density (kg/m^3)	Young's modulus (GPa)	Poisson's ratio	Yield stress (MPa)	Fracture strain	Stress at fracture (MPa)
Pin	7850	200	0.3	600	0.14	827
L-beam	7850	200	0.3	344	0.16	448
Flat plate	7850	200	0.3	344	0.16	448

interference pin neck contains 30 teeth on its surface, so the sections were modeled using a 12-degree cut of the interference pin, L-beam, and flat plate. A friction coefficient of 0.2 was used for contacts among the interference pin, L-beam, and flat plate sections. Plastic behaviors were enabled for the interference pin, L-beam, and flat plate sections so that plastic deformation of the installation process could be modeled. The interference pin, L-beam, and flat plate sections were meshed such that meshes near the interference surfaces were finer than those further away. The material properties used in the simulation are given in Table 1.

The interference pin, L-beam, and flat plate sections assembly is shown in Figure 3. The bottom surface of the interference pin section was positioned above the top surface of the L-beam section. To represent the effects of the surrounding materials on the sections, fixed boundary conditions were placed on the sectioned surfaces to restrict displacements normal to those surfaces. FE model simulation of the interference pin installation consisted of two steps. In the first step, the interference pin section was driven into the L-beam and flat plate sections by displacing elements of the interference pin section downward by a distance equal to the combined thickness of the L-beam and flat plate sections via a displacement constraint, as shown in Figure 3(a). For this step, a fixed boundary condition was applied to the lower surface of the flat plate section to simulate an actual interference pin installation, where the lower surface of the flat plate would be fixed. In the second step, a clamping stress of 150 MPa was applied to the L-beam and flat plate sections, as shown in Figure 3(b). The clamping stress acted over an area equal to the area of the interference pin head that was in contact with the L-beam, and it was applied for 9.6 s to allow time for the stress and deformation to converge. In this step, a fixed boundary condition was not applied to the lower surface of the flat plate section, but fixed boundary conditions were applied to the upper and lower surfaces of the interference pin section to simulate an actual interference pin fixed in a hole via an interference fit. During the clamping step, as expected, it was observed that the simulation was sensitive to the clamping stress applied. For example, clamping stresses less than approximately 50 MPa resulted in incomplete contact between the L-beam and flat plate sections, while clamping stresses greater than approximately 50 MPa resulted in contact stress propagation in the contact region, where the contact region area increased with the increasing clamping stress.

After completion of the clamping step, the area and stiffness of the contact region could be determined. The L-beam and flat plate sections were in contact due to the applied clamping stress, where stress propagated on those surfaces as shown in Figure 4. The area of the contact region is the area of the L-beam and flat plate section surfaces with non-zero contact stresses. Because clamping is a semi-static process, the area of the contact region fluctuated with time before converging to a constant value. The converged radius of the contact region was obtained from convergence of the FE model data over the 9.6 s duration. From convergence of the data, the total distance from the center of the interference pin to the propagated edge of the contact region was 0.0278 m, which was the value assigned to the radius of the cylinder.

To determine the Young's modulus of the cylinder, the deformation parallel to the direction of the applied clamping stress was found by capturing the combined thickness of the L-beam and flat plate sections after convergence of the FE model data over the 9.6 s duration. By subtracting the converged combined thickness due to clamping from the original combined thickness prior to clamping, the deformation was calculated to be 1.28×10^{-5} m. The Young's modulus of the L-beam and flat plate sections within the contact region was calculated from the stress-strain relation for a material within the elastic region

$$E_p = \frac{FL}{A_p \Delta L} \quad (1)$$

where E_p is the Young's modulus of the L-beam and flat plate sections within the contact region, F is the clamping force on the L-beam and flat plate sections, L is the original combined thickness of the L-beam and flat plate sections, A_p is the area of the L-beam and flat plate contact region sections, and ΔL is the deformation, or the change in the combined thickness of the L-beam and flat plate sections after convergence of the FE model data due to the applied clamping stress. Variable A_p is calculated by subtracting the area of the interference pin section from the combined area of the converged contact region section and the interference pin section

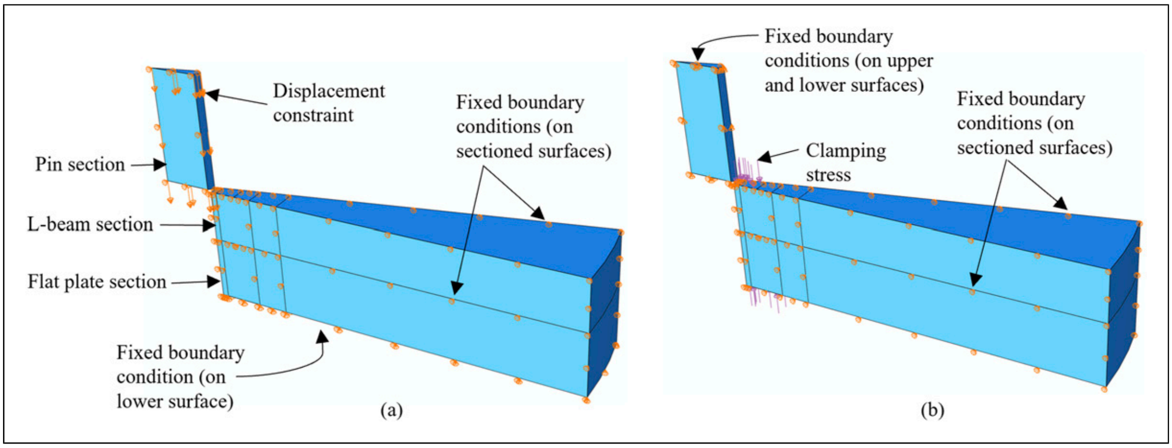

Figure 3. (a) FE model of the interference pin, L-beam, and flat plate sections assembly with a displacement constraint on the interference pin section, fixed boundary conditions on the sectioned surfaces, and a fixed boundary condition on the lower surface of the flat plate section during the interference pin installation step. (b) The assembly with an applied clamping stress, fixed boundary conditions on the sectioned surfaces, and fixed boundary conditions on the upper and lower surfaces of the interference pin section during the clamping step.

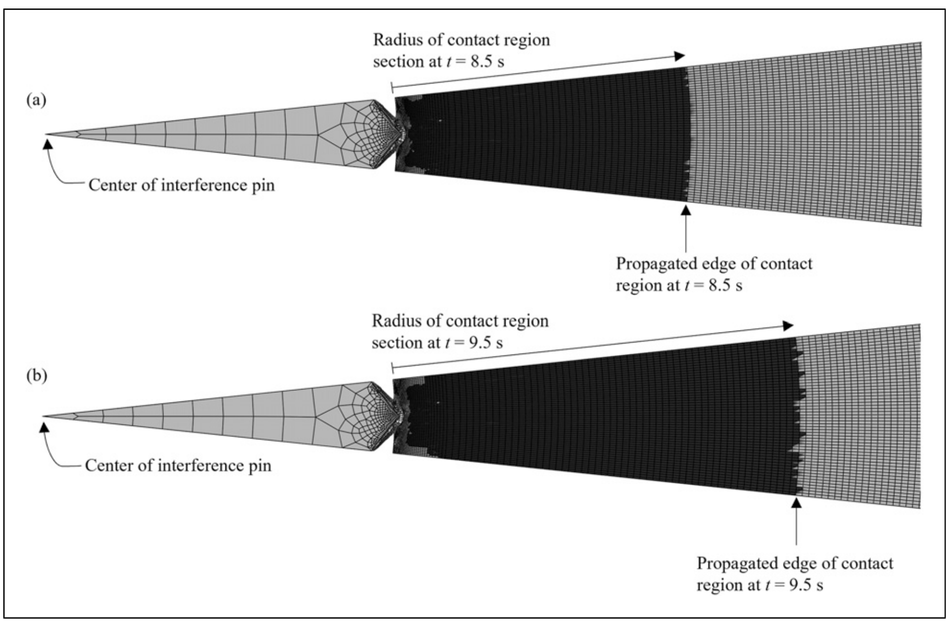

Figure 4. Contact regions indicated by dark-shaded regions at different times, t: (a) 8.5 s and (b) 9.5 s; non-contact regions are shown as light-shaded regions.

$$A_p = \frac{\pi}{30} \times \left(0.0278^2 - 0.0135^2\right) \mathrm{m}^2 = 6.18 \times 10^{-5} \mathrm{m}^2 \quad (2)$$

When values of the four variables were substituted into equation (1), the Young's modulus of the L-beam and flat plate sections was calculated to be 135 GPa. The calculated value shows that the Young's modulus of the L-beam and flat plate sections within the contact region was reduced due to clamping. While the Young's modulus of the L-beam and flat plate sections within the contact region changed, the Young's modulus of the interference pin remained unchanged, as it did not undergo clamping. The Young's modulus of the cylinder was calculated using a weighted average of Young's modulus values of the interference pin, L-beam, and flat plate sections

$$E_c = \frac{E_p A_p + E_i A_i}{A_p + A_i} \quad (3)$$

where E_c is the Young's modulus of the cylinder, E_i is the Young's modulus of the interference pin section, and A_i is the area of the interference pin section. The area of the interference pin section was calculated by

$$A_i = \frac{\pi}{30} \times (0.0135\,\mathrm{m})^2 = 1.91 \times 10^{-5}\,\mathrm{m}^2 \quad (4)$$

By using equation (3), the Young's modulus of the cylinder was calculated to be 150 GPa. With estimated values for the radius and Young's modulus of the cylinder, the cylinder could then be used to model the interference pin connections in the FE model of the lattice tower component.

2.2. Updating Young's modulus values of the L-beam and flat plate in the FE model

Modal testing was performed on the L-beam and flat plate, shown in Figure 2, to obtain their experimental natural frequencies. Reflective tape was attached to the surfaces of the L-beam and flat plate to maximize backscattering of laser light. Strings were used to hang the L-beam and flat plate to yield free boundary conditions. The suspended L-beam and flat plate were each excited by an MB Dynamics MODAL-50 shaker with a periodic chirp signal with a frequency range of 0–5000 Hz. A Polytec PSV-500-3D SLDV was used to capture 3D responses of the L-beam and flat plate. Frequency spectra and identified natural frequencies of the L-beam and flat plate are shown in Figures 5(a) and (b), respectively, where frequency spectra are plotted as solid black lines and natural frequencies of elastic modes are marked by red circles. Seven natural frequencies of the L-beam and nine natural frequencies of the flat plate were identified in the tested frequency range of 0–5000 Hz.

FE models were created for the L-beam and flat plate, where the L-beam was meshed using C3D4 tetrahedrons due to its fillet, while the flat plate was meshed using C3D8R hexahedrons. Element sizes were determined by convergence of the FE modal analysis results as element size decreased. The mesh density was sufficient to account for any inaccuracies introduced by using C3D4 tetrahedrons. Both models were given free boundary conditions to simulate the free boundary conditions of the experimental test specimens, and nominal values for the density and Poisson's ratio of the L-beam and flat plate, as reported in Table 1, were used in the FE model. For calibration of the L-beam and flat plate FE models to the experimental test specimens, the L-beam and flat plate models were initially assigned nominal Young's modulus values of 200 GPa, as reported in Table 1. A manual model updating procedure was performed where the Young's modulus values of the L-beam and flat plate were iteratively varied to minimize the maximum percent difference between the theoretical and experimental natural frequencies. The percent differences between the theoretical and experimental natural frequencies were calculated by

$$\text{Percent Difference}\,(\%) = \frac{|x-y|}{[(x+y)/2]} \times 100 \quad (5)$$

where x is the natural frequency in the first data set and y is the corresponding natural frequency in the second data set. In this work, where only a single parameter was optimized, manual model updating was sufficient. However, in a case where there is a requirement for optimization of many parameters, a surrogate model may be employed (Salachoris et al., 2023). In cases of FE modeling, a surrogate model is a lower fidelity statistical model representing a higher fidelity FE model (Kudela and Matousek, 2022) whose purpose is to solve complex optimization problems while requiring less computational power (Salachoris et al., 2023; Standoli et al., 2021). Through manual model updating in this work, the L-beam and flat plate were estimated to have Young's modulus values of 205 GPa and 206 GPa, respectively. The maximum percent difference between the theoretical and experimental natural frequencies of the L-beam is 0.69% while that of the flat plate is 1.10%, as shown in Tables 2 and 3, respectively.

2.3. Developing the FE model of the lattice tower component

The FE models of the L-beam and flat plate, along with the cylinder, were used to create the FE assembly model of the lattice tower component. A modification was made to the L-beam and flat plate models by making the radius of the fastener holes equal to that of the cylinder. The assembled FE model of the lattice tower component, including the interference pins, nuts, and washers, is shown in Figure 6. Tie constraints were used to attach four cylinders to the L-beam and flat plates. Frictionless contacts were enabled between surfaces of the L-beam and flat plates, and the assembly was given free boundary conditions. The complete assembly was meshed using an element size of 0.002 m, as that is the mesh size where results converged. For example, reducing the element size to 0.0015 m results in a natural frequency percent difference of less than 1% when compared with the natural frequency results found using an element size of 0.002 m. An FE modal analysis simulation was performed and ten out-of-plane elastic modes and one in-plane elastic mode were observed within a 0–2000 Hz frequency range.

2.4. Modal testing of the lattice tower component

A modal shaker test of the lattice tower component was conducted to obtain its experimental natural frequencies and mode shapes. The lattice tower component was suspended by a string, as shown in Figure 7, to yield free boundary conditions. The surface of the lattice tower component that was normal to the laser of the middle 3D SLDV laser head is referred to as the front surface, while the surface parallel to it is referred to as the side surface, as shown in Figure 8. The mirror was used to reflect the lasers from the three laser heads onto the side surface to

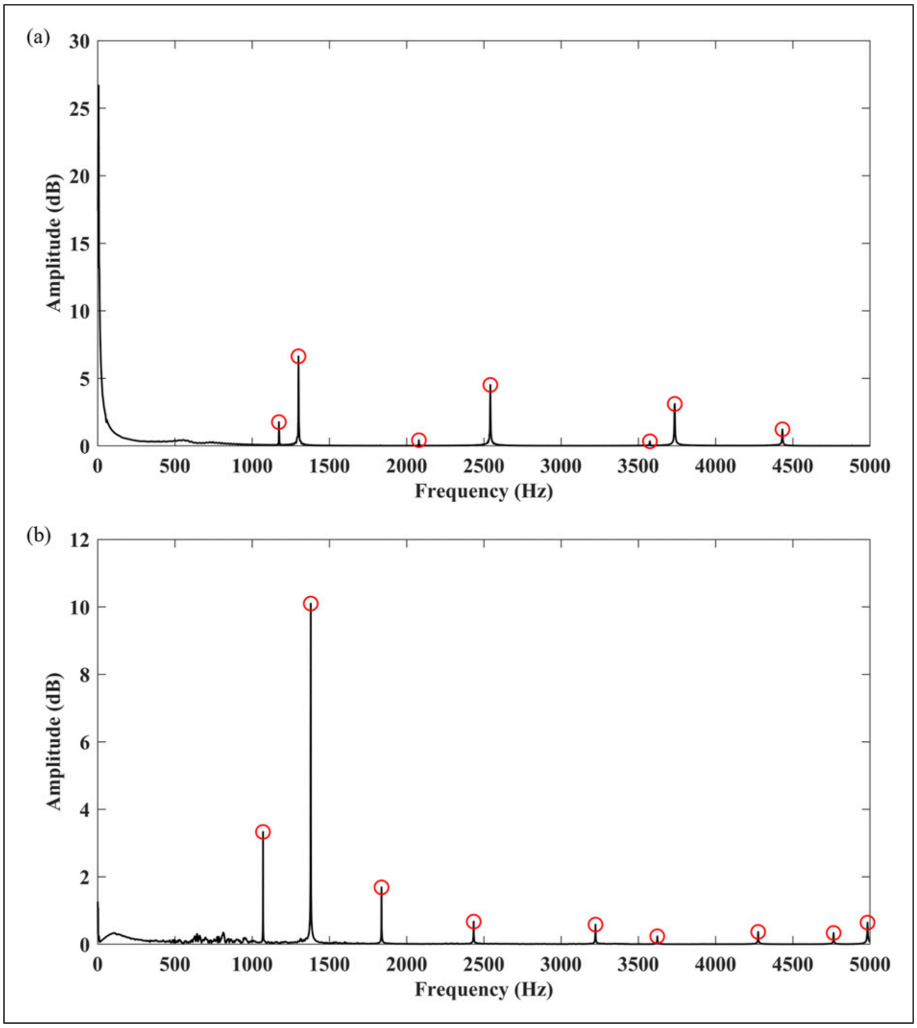

Figure 5. Frequency spectra of the (a) L-beam and (b) flat plate plotted as solid black lines with identified natural frequencies of elastic modes marked by red circles.

Table 2. Comparison of the L-beam's first seven natural frequencies captured from the updated FE model with those from the modal test.

Mode no.	FE model (Hz)	Modal test (Hz)	Percent difference (%)
1	1167	1173	0.51
2	1293	1300	0.54
3	2075	2080	0.24
4	2539	2542	0.12
5	3573	3575	0.06
6	3762	3736	0.69
7	4441	4433	0.18

Table 3. Comparison of the flat plate's first eight natural frequencies captured from the updated FE model with those from the modal test.

Mode no.	FE model (Hz)	Modal test (Hz)	Percent difference (%)
1	1072	1070	0.19
2	1380	1380	0.00
3	1823	1838	0.82
4	2420	2434	0.58
5	3230	3223	0.22
6	3657	3623	0.93
7	4246	4276	0.70
8	4712	4764	1.10

measure the response of the side surface (Yuan and Zhu, 2021, 2024). Calibration of the 3D SLDV was conducted, and grids of measurement points on the front and side surfaces of the component were designed, as shown in Figure 8. There were 156 measurement points on the front surface and 81 measurement points on the side surface. The coordinates of the measurement points on the front and side surfaces were captured using the global coordinate system shown in Figure 7(b), where the X axis is along the in-plane horizontal direction, Y axis is along the in-plane vertical direction, and Z axis is along the out-of-plane direction. Eleven elastic modes were found within the 0-2000 Hz frequency range, as shown in Figure 9. The excitation of the lattice tower component was in the Z direction for the ten out-of-plane modes and in the X direction for the single in-plane mode.

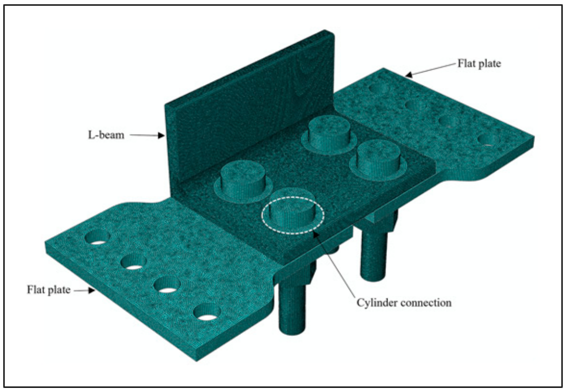

Figure 6. FE model of the lattice tower component.

3. Results and analysis

The natural frequencies of the first 11 elastic modes determined by the FE model and modal test, and their percent differences, are presented in Table 4. The maximum percent difference is 3.21%, demonstrating good agreement between the theoretical and experimental results.

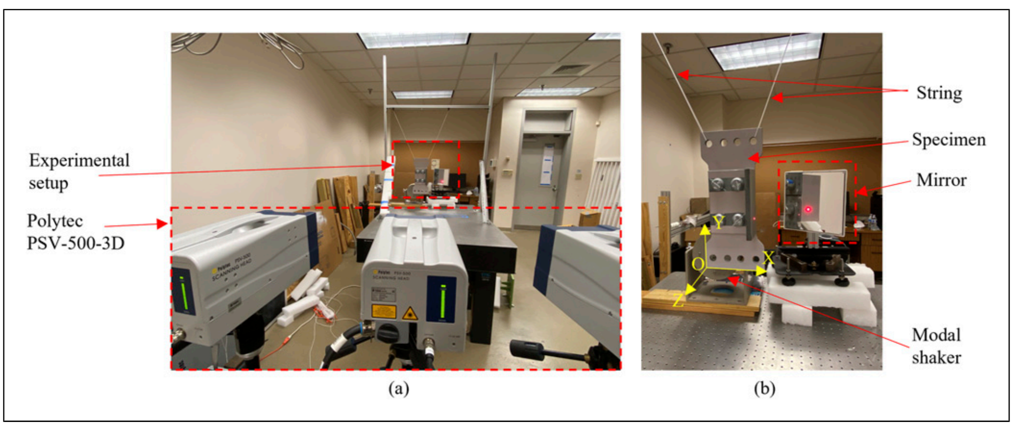

Figure 7. (a) Experimental setup with the Polytec PSV-500-3D SLDV and (b) a close-up view of the experimental setup with the coordinate system, the MB Dynamics MODAL-50 shaker, and a mirror.

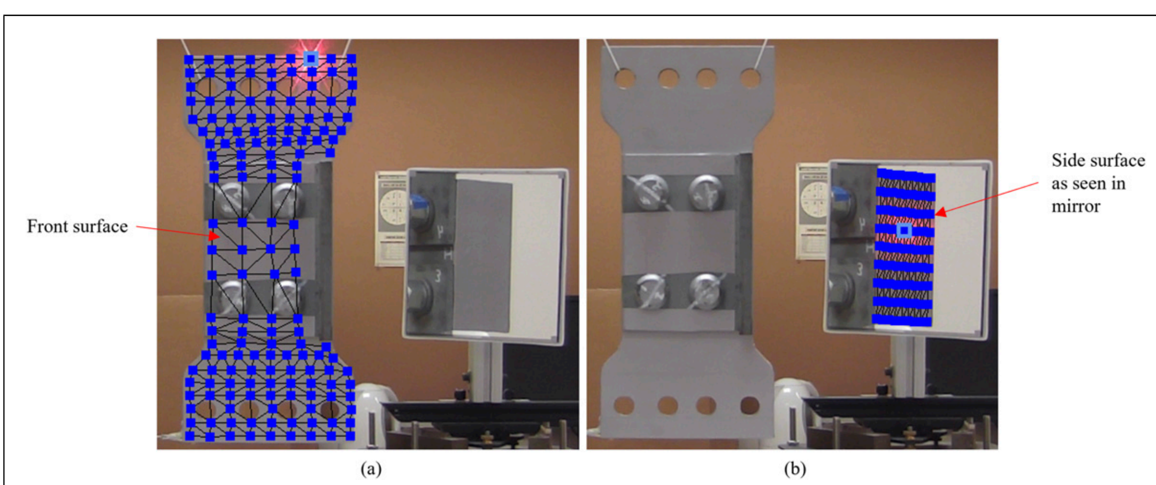

Figure 8. Grids of measurement points on the (a) front surface and (b) side surface of the lattice tower component.

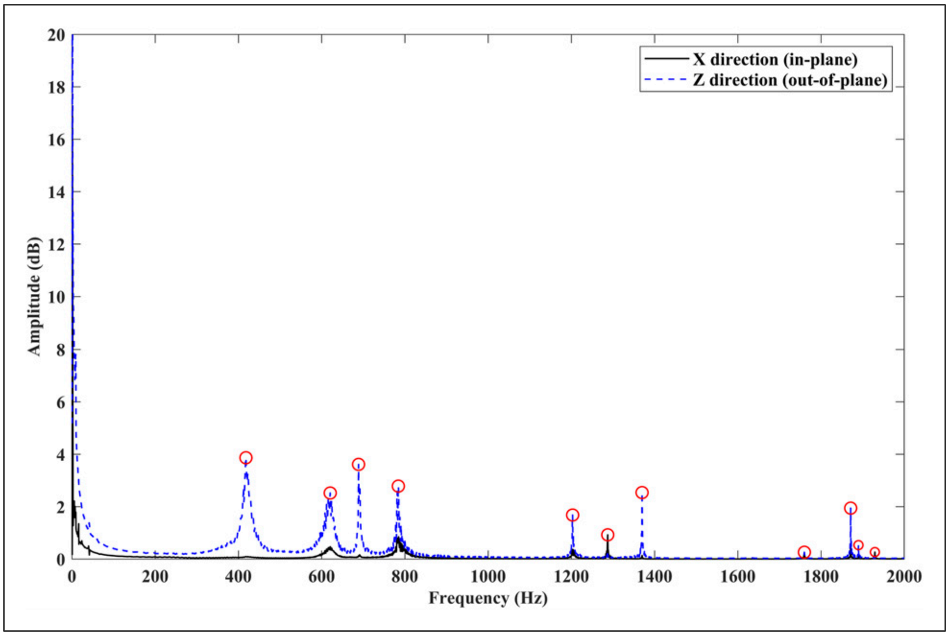

Figure 9. Frequency spectra of the lattice tower component, where the response spectrum along the X direction is plotted as a solid black line, the response spectrum along the Z direction is plotted as a dashed blue line, and natural frequencies are indicated by red circles.

Table 4. FE model and modal test natural frequencies of the first 11 modes, and their percent differences.

Mode no.	FE model (Hz)	Modal test (Hz)	Percent difference (%)
1	422	429	1.65
2	629	622	1.12
3	678	691	1.90
4	805	796	1.12
5	1221	1203	1.49
6	1328	1286	3.21
7	1388	1369	1.38
8	1815	1759	3.13
9	1876	1870	0.32
10	1903	1890	0.69
11	1983	1932	2.61

The theoretical and experimental mode shapes are presented in Figures 10(a) and (b), respectively, and were compared using a MAC value calculation, which provides a measure of correlation between two mode shapes, given by

$$\mathrm{MAC} = \frac{\left[\sum_{j=1}^{n}\{U_{ej}\}^{H}\{U_{tj}\}\right]^{2}}{\left(\sum_{j=1}^{n}(\{U_{ej}\}^{H}\{U_{ej}\})\right)\left(\sum_{j=1}^{n}(\{U_{tj}\}^{H}\{U_{tj}\})\right)} \quad (6)$$

where H denotes the Hermitian transpose, n is the number of measurement points, and U_{ej} and U_{tj} are components of displacement vectors at the measurement point j for experimental and theoretical mode shapes, respectively (Ewins, 2000). MAC values range from 0 to 1, where a value close to 0 indicates low correlation and a value close to 1 indicates high correlation. Modes 1 through 10 are out-of-plane modes, and only front surface out-of-plane displacements were used in the MAC value calculations. Mode 11 is an in-plane mode, and it was the only mode where front surface in-plane displacements were used in the MAC value calculations. All modes of the side surface are out-of-plane modes relative to the side surface itself, and all points on the side surface had their out-of-plane displacements, relative to the side surface itself, used in the MAC value calculations.

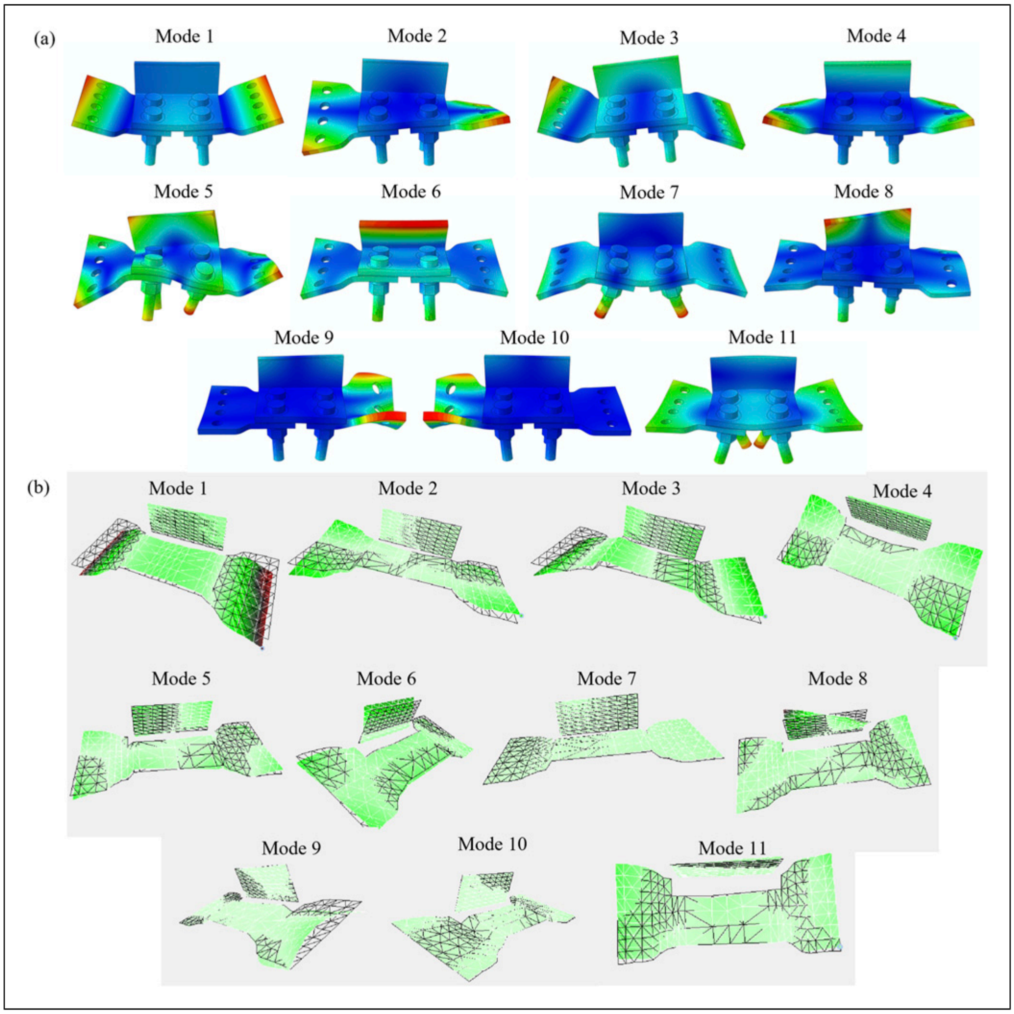

Figure 10. First 11 (a) FE model and (b) modal test elastic mode shapes of the lattice tower component.

Table 5. MAC values for the first 11 elastic mode shapes of the lattice tower component, where, for example, FE1 and MT1 correspond to the first FE model and modal test mode shapes, respectively.

	MT1	MT2	MT3	MT4	MT5	MT6	MT7	MT8	MT9	MT10	MT11
FE1	0.98	0.00	0.01	0.01	0.00	0.01	0.16	0.00	0.00	0.01	0.01
FE2	0.00	0.98	0.02	0.00	0.30	0.00	0.00	0.02	0.04	0.00	0.01
FE3	0.00	0.05	0.99	0.00	0.00	0.00	0.01	0.01	0.00	0.00	0.03
FE4	0.00	0.00	0.00	0.99	0.01	0.00	0.03	0.00	0.02	0.00	0.10
FE5	0.00	0.26	0.00	0.01	0.97	0.00	0.01	0.17	0.03	0.02	0.02
FE6	0.00	0.00	0.00	0.01	0.00	0.97	0.02	0.00	0.00	0.00	0.00
FE7	0.12	0.01	0.02	0.03	0.03	0.06	0.92	0.00	0.08	0.01	0.05
FE8	0.00	0.02	0.00	0.00	0.15	0.00	0.00	0.95	0.00	0.01	0.02
FE9	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.01	0.96	0.04	0.04
FE10	0.00	0.00	0.00	0.00	0.01	0.00	0.02	0.00	0.09	0.93	0.04
FE11	0.29	0.00	0.01	0.08	0.00	0.00	0.11	0.00	0.03	0.00	0.98

All 237 measurement points taken during the modal test were used to perform the MAC value calculations. The MAC values are presented in Table 5, where values on the diagonal are 0.92 or greater, while off-diagonal values are 0.30 or less, indicating good correlation between the FE model and modal test mode shapes.

Modes 2 and 5 have relatively high off-diagonal MAC values of 0.26 and 0.30 due to the flat plates having similar torsional bending in those modes. Note that 128 measurement points were used to measure the responses of the flat plates, while only 28 measurement points were used to measure the response of the front surface of the L-beam. More measurement points on the front surface of the L-beam could better distinguish these two mode shapes and reduce their MAC values. For modes 1 and 11, the (11, 1) entry has a value of 0.29, which is significantly greater than the (1, 11) entry which has a value of 0.01. From examining the mode shapes in Figure 10, theoretical mode 11 has strong in-plane characteristics with minor out-of-plane characteristics. Experimental mode 1 has out-of-plane characteristics like theoretical mode 11, yielding the MAC value of 0.29. However, theoretical mode 1 has strong out-of-plane characteristics and experimental mode 11 has strong in-plane characteristics due to the in-plane excitation, yielding the low MAC value of 0.01.

4. Conclusion

The purpose of this research was to develop a novel FE modeling method for accurate dynamic analysis of a wind turbine lattice tower component with interference pin connections. First, a two-step modeling process simulated the installation of an interference pin into an L-beam and flat plate, with subsequent clamping of the L-beam and flat plate by a nut to obtain the radius and Young's modulus values of a cylinder. Next, the FE models of the L-beam and flat plate were calibrated to the experimental test specimens through model updating of the Young's modulus values. An FE model of the lattice tower component was then created using the cylinders in place of the interference pin connections, which modeled the dynamic behaviors of the interference pin connections. An FE modal analysis simulation was performed, where ten out-of-plane modes and one in-plane mode were identified. A modal shaker test involving a 3D SLDV and a mirror was subsequently used to obtain the out-of-plane and in-plane frequency spectra and mode shapes of the lattice tower component. The FE model predicted the first 11 natural frequencies and mode shapes with a good degree of accuracy when compared to the results from the modal test, where the maximum percent difference between the theoretical and experimental natural frequencies of the component was 3.21%, and the MAC values between the theoretical and experimental mode shapes were 0.92 or greater.

Designing a simplified FE model with sections of the interference pin, L-beam, and flat plate to reduce the computation cost was critical to avoid exceeding computer memory constraints. When simulating the interference pin installation step, fine meshes were needed to accurately capture material stress and deformation, where an increase in the FE model mesh density results in increased computer memory requirements. During the model calibration step of the L-beam and flat plate, only the Young's modulus values were updated. To further reduce the maximum percent difference between the theoretical and experimental natural frequencies and to improve the MAC values between the theoretical and experimental mode shapes, additional parameters such as the material density and Poisson's ratio could be optimized. To facilitate this optimization procedure, a surrogate model can be employed for parameter optimization in lieu of a manual model updating procedure where parameters are iteratively varied. Ultimately, the methodology developed in this work for modeling interference pin connections using cylinders was shown to be an effective technique to account for the local stiffness changes inherent in interference fits. It was shown that the FE model simulation results computed using the simplified model could be readily applied to the full model of the wind turbine lattice tower component. In future research, the results of the wind turbine lattice tower component model can be integrated into the complete wind turbine lattice tower structure. A subsequent experimental modal analysis of the wind turbine lattice tower can be performed to compare the theoretical results to the experimental results.

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Data availability statement

Available upon request.

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